

MARK LORENZ YAMANAKA

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SUMMARY

Licensed Aeronautical Engineer (PRC, Philippines) with project experience in aircraft structure and mechanical design, carrying a baseline from concept and CAD through stress analysis, load cases, and manufacturing drawings (SolidWorks, CATIA V6, GD&T to ASME Y14.5-2018). Rotorcraft design coursework and a two-blade rotor design project; aircraft maintenance internship on Airbus and ATR fleets under CAAP airworthiness rules. Seeking a Structure and Mechanical Design Engineer role supporting certifiable helicopter modifications under CS-27 and CS-29.

SKILLS

Design & CAD: SolidWorks, CATIA V6; GD&T and drawing packages to ASME Y14.5-2018

Analysis & FEA: ANSYS Workbench and Mechanical, SpaceClaim; CalculiX (PrePoMax); ANSYS Fluent (CFD)

Standards & Methods: FAR Part 27/29 (covered in the PRC Aeronautical Engineering licensure exam); Bruhn airframe stress methods; familiarity with the helicopter airworthiness basis from rotorcraft design coursework

Programming & Tools: Python, Git/GitHub, Excel, XFOIL, Asana

Languages: English (Native), Filipino (Professional)

Certifications: PRC Professional Aeronautical Engineer License (Philippines, 2024)

PROJECTS

Aircraft Design Project: MK-10 | github.com/dmmarkwork-cloud/mk10-light-aircraft-design

- Designed a two-seat FAR Part 23 light aircraft (1,649 lb gross weight) from concept through structure: sized wing loading and power-to-weight from mission and constraint analysis, verified climb, ceiling, and range, built a V-n flight envelope, and sized front and rear wing spars from a 10-strip spanwise load distribution (per Bruhn) to a 4.4G limit and 7.4G ultimate; resolved the design into control surfaces, landing-gear load cases, and a SolidWorks three-view.

Two-Blade Helicopter Rotor Design | github.com/dmmarkwork-cloud/rotor-blade-helicopter-design

- Took a two-blade rotor from blade element theory to a dimensioned CAD assembly (71.1 in diameter, nonlinear twist from 23.5 to 78.0 degrees across three stations); modelled blade and hub parts, built the master assembly with a four-bolt hub pattern, and produced a formal three-view drawing in SolidWorks.

SS316 Conical Convergent-Divergent Nozzle | github.com/dmmarkwork-cloud/ss316-cd-nozzle-feasibility

- Ran coupled CFD-FEA (one-way FSI) on an uncooled SS316 nozzle ($P_c = 2$ MPa, $T_c = 800$ K), converging the flow solution to 0.00003% mass imbalance and confirming $FoS = 7.12$ at the governing chamber-to-converging-section junction under combined thermal and pressure load; delivered a design-for-manufacture package in CATIA V6 (GD&T drawings to ASME Y14.5-2018), with a worst-case tolerance stack-up and a bolted joint (8x M8 A286) sized to spec.

EDUCATION

Bachelor of Science in Aeronautical Engineering

Jul 2020 – May 2024

Holy Angel University

- Honours: Dean's List (GPA: 3.6/4.0)
- Relevant Coursework: Aircraft Structures; Design and System (Aircraft, Helicopter, UAV); Aircraft Systems; Aerodynamics; Powerplant; Reliability Engineering; Engineering Data Analysis

EXPERIENCE

Aircraft Maintenance Engineer Intern | Dornier Technology Inc.

May 2023 – Jul 2023

- Supported 10+ base and line checks under PCAR Part 6 (modelled on FAR Part 145) across four aircraft types (A320, B737, ATR 72, BAe 146), seeing at first hand how a type-certified airframe is inspected, maintained, and released to service.
- Worked from customer task/job cards on access panel removal and refit, fastener removal and installation, and visual inspection for structural damage and corrosion, with hands-on exposure to dye penetrant and borescope inspection.
- Carried out 15+ pre- and post-inspections of airframe structure, systems, and installed equipment; recorded condition findings and defects for the responsible engineer to review and clear.
- Prepared work orders and maintenance records for operators including Cebu Pacific and Royal Air, supporting each certificate for return to service (RTS), and helped ready documentation for an internal audit.

Part-Time Leader / Sub-Leader, Packaging and Logistics | ZOZOBASE

Feb 2018 – May 2026

- Organised and monitored shift activities for a 15-member operations team, maintaining accuracy and schedule adherence during peak demand.
- Coordinated shift teams in an operational environment, briefing floor operators on process and priority changes and reporting daily to shift supervisors.
- Proposed workflow changes that raised throughput 20% and cut cycle time 12%, later adopted by shift supervisors.